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Revised Simplex Method

Standard form-I

(Identity matrix is obtained after introducing slack variables only)

$$\text{Max } Z = x_1 + 2x_2$$

$$\text{Subject to } \begin{cases} x_1 + x_2 \leq 3 \\ x_1 + 2x_2 \leq 5 \\ 3x_1 + x_2 \leq 6 \\ x_1, x_2 \geq 0 \end{cases}$$

$$\begin{cases} x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 + x_4 = 5 \\ 3x_1 + x_2 + x_5 = 6 \end{cases}$$

$x_1, x_2, x_3, x_4, x_5 \geq 0$
 $x_3, x_4, x_5 \rightarrow$ slack variables

Step-I : (i) All $b_i \geq 0$

(ii) Objective f_ns is max or not

(iii) convert each inequalities to eq_n.

(iv) treat the objective f_n as 1st constraint eq₀

$$Z - x_1 - 2x_2 = 0 \quad \text{--- (1)}$$

$$\begin{cases} x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 + x_4 = 5 \\ 3x_1 + x_2 + x_5 = 6 \end{cases}$$

Step-II: We will represent the constraints into matrix form.

$$\begin{matrix} & z & x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & -1 & -2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 3 & 1 & 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} Z \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} & = & \begin{bmatrix} 0 \\ 3 \\ 5 \\ 6 \end{bmatrix} \end{matrix}$$

$\downarrow$ B_0 $\underbrace{\quad}_a$ $\downarrow$ B_1 $\downarrow$ B_2 $\downarrow$ B_3

Basis Matrix $\rightarrow$

$$B = [B_0, B_1, B_2, B_3]$$

$$a = [x_1, x_2] = [a_1^{(1)}, a_2^{(1)}]$$

Initial Revised Simplex Table-I

Basic variables	B^{-1}				$x_B^{(1)}$	$x_K^{(1)}$	Min (x_B/x_K)
	$B_0^{(1)}$	$B_1^{(1)}$	$B_2^{(1)}$	$B_3^{(1)}$			
Z	1	0	0	0	0	-2	—
x_3	0	1	0	0	3	1	$3/1=3$
x_4	0	0	1	0	5	2	$5/2=2.5$
x_5	0	0	0	1	6	1	$6/1=6$

Additional Table.

$a_1^{(1)}$	$a_2^{(1)}$
-1	-2
1	1
1	2
3	1

Step-III: ① Check the optimality condition.

$$\Delta_j \geq 0$$

$$\Delta_j = [\text{first row of } B^{-1}] \times [a_1^{(1)}, a_2^{(1)}]$$

$$= (1, 0, 0, 0) \times \begin{bmatrix} -1 & -2 \\ 1 & 1 \\ 1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{pmatrix} -1 & -2 \\ \Delta_1 & \Delta_2 \end{pmatrix}$$

$$\left. \begin{array}{l} \Delta_1 = -1 \\ \Delta_2 = -2 \end{array} \right\}$$

As not all $\Delta_j \geq 0 \quad \forall j = \{1, 2\}$,
therefore, optimality is not arise.

② To find entering and leaving variables

$$\text{Min}(\Delta_1, \Delta_2) = \Delta_2$$

$$\Delta_2 = -2$$

$$K = 2$$

a_2 is entering variable.

leaving variable = x_4

2 $\rightarrow$ key element.

③ Intermediate co-efficient.

B_1	B_2	B_3	x_B	x_K
0	0	0	0	-2
1	0	0	3	1
0	1	0	5	2
0	0	1	6	1

B_1	B_2	B_3	x_B	x_K
0	1	0	5	0
1	$-\frac{1}{2}$	0	$\frac{1}{2}$	0
0	$\frac{1}{2}$	0	$\frac{7}{2}$	1
0	$\frac{7}{2}$	1	$\frac{7}{2}$	0

$$R_1 \leftarrow R_1 + 2R_3$$

$$R_2 \leftarrow R_2 - R_3$$

$$R_3 \leftarrow \frac{1}{2}R_3$$

$$R_4 \leftarrow R_4 - R_3$$

Revised Simplex Table - II

B-V	B_2^{-1}				x_B	x_K	Min
	$B_1^{(2)}$	$B_2^{(2)}$	$B_3^{(2)}$	$B_4^{(2)}$			
Z	1	0	1	0	5		
x_3	0	1	$-1/2$	0	$1/2$		
x_2	0	0	$1/2$	0	$5/2$		
x_1	0	0	$1/2$	1	$7/2$		

$a_1^{(2)}$	$a_2^{(2)}$
-1	0
1	0
1	1
0	0

optimality condition.

$$\Delta_j = [\text{first row of } B^{-1}] \times \text{Addition Table.}$$

$$= [0, 1]$$

$$\Delta_j \geq 0$$

Soln

$$Z = 5,$$

$$x_1 = 0, x_2 = 5/2$$